

Fourier columns and the inner-carrier obstruction

Free flip on C^2

$G = \mathbb{Z}/2\mathbb{Z}$; s swaps the two coordinates

$\text{diag}(C^2)$ and $U = [[0,1],[1,0]]$ generate M_2

$x = \text{diag}(1,2) + \text{diag}(3,4)$ $U = [[1,3],[4,2]]$

$E(x^* x) = \text{diag}(17,13)$

$E(x x^*) = \text{diag}(10,20)$

Inner action on M_2

$W = \text{diag}(1,-1)$; $\alpha s = \text{Ad } W$

$v = \pi(W)^*$ is a central involution

$v^2 = 1$; v is not scalar (its s coefficient $\neq 0$)

Crossed product = $M_2 \otimes C^2$

The central projections are $(1+v)/2$ and $(1-v)/2$

The exact positive tail

Keep $F = \{e\}$: $SF(x) = \text{diag}(1,2)$

$E((x - SF(x))^* (x - SF(x))) = \alpha s(\text{diag}(9,16)) = \text{diag}(16,9)$

For $\phi(\text{diag}(d_1, d_2)) = (d_1 + d_2)/2$, squared Hilbert error = $25/2$.

The proved general limit is the normal Hilbert-seminorm limit (FC5-FC6).

This finite computation makes no claim about raw strong* Fourier convergence.